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Human DNA Profiling

DNA Fingerprinting

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What is DNA Fingerprinting

- The chemical structure of everyone's DNA is the same.
- The only difference between people (or any animal) is the order or sequence of the base pairs.
- There are so many millions of base pairs in each person's DNA that every person has a different sequence.
- Using these sequences, every person could be identified solely by the sequence of their base pairs.

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What is DNA Fingerprinting?

- However, because there are so many millions of base pairs, the task would be very time-consuming.
- Instead, scientists are able to use a shorter method, because of repeating patterns in DNA called VNTRs which are known to vary among individuals a great deal
- These patterns do not, however, give an individual "fingerprint," but they are able to determine whether two DNA samples are from the same person, related people, or non-related people.

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How is it done?

1. Performing a Southern Blot
2. Making a Radioactive Probe
3. Creating a Hybridization Reaction

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Southern Blot

- The Southern Blot is one way to analyze the genetic patterns which appear in a person's DNA. It involves…

1. **Isolating the DNA from rest of the cell:**
   - This can be done either chemically, by using a detergent to wash the extra material from the DNA, or mechanically, by applying a large amount of pressure in order to "squeeze out" the DNA.

2. **Denaturing the DNA:**
   - To render the entire DNA as a single strand.
   - This can be done either by heating or chemically treating the DNA in the gel.

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Southern Blot

3. **Cutting the DNA into several pieces of different sizes**.
   - This is done using one or more restriction enzymes.

4. **Sorting the DNA pieces by size – Size Fractionation**
   - The process by which the size separation is done is called **gel electrophoresis.**
   - The DNA is poured into a gel, such as agarose, and an electrical charge is applied to the gel, with the positive & negative charges
   - Because DNA has a slightly negative charge, the pieces of DNA will be attracted towards positive charge
   - The smaller pieces, however, will be able to move more quickly and thus further towards the positive charge

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Southern Blot

5. Blotting the DNA:

- The gel with the size-fractionated DNA is applied to a sheet of nitrocellulose paper, and then baked to permanently attach the DNA to the sheet.
- The Southern Blot is now ready to be analyzed.

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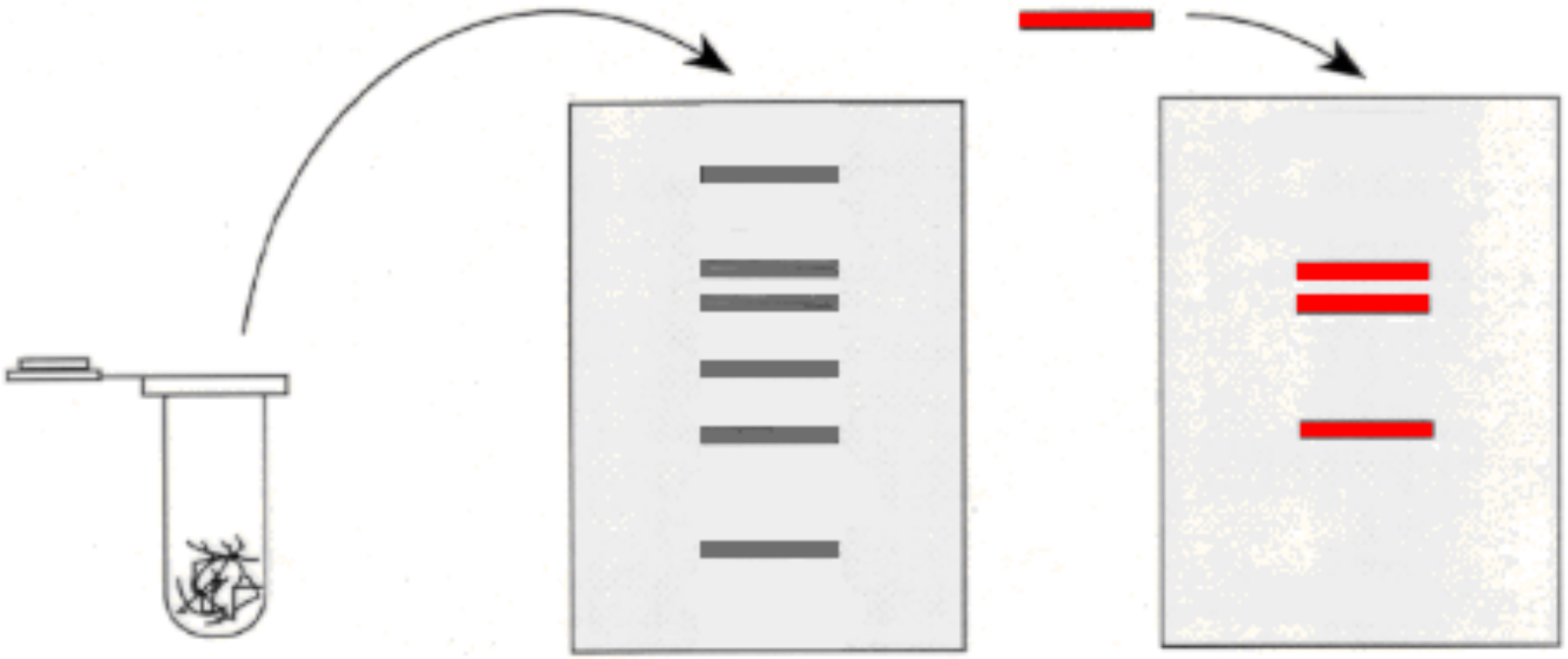
Radioactive Probe

- In order to analyze a Southern Blot, a radioactive genetic probe is used in a hybridization reaction with the DNA in question.
- If an X-ray is taken of the Southern Blot after a radioactive probe has been allowed to bond with the denatured DNA on the paper, only the areas where the radioactive probe binds will show up on the film.
- This allows researchers to identify, in a particular person's DNA, the occurrence and frequency of the particular genetic pattern contained in the probe.

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Southern Blot & Probing

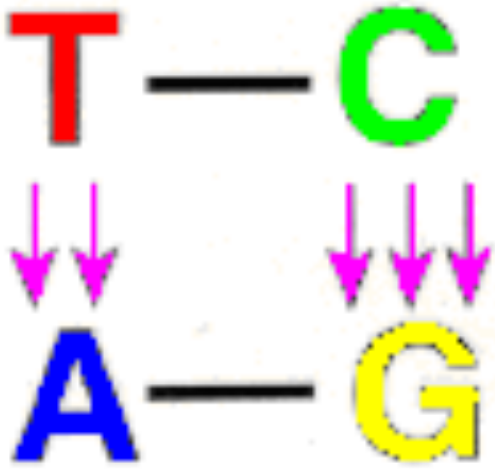


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Hybridization

- Hybridization is the coming together, or binding, of two genetic sequences.
- The binding occurs because of the hydrogen bonds [pink] between base pairs.
- Between a A base and a T base, there are two hydrogen bonds; between a C base and a G base, there are three hydrogen bonds.

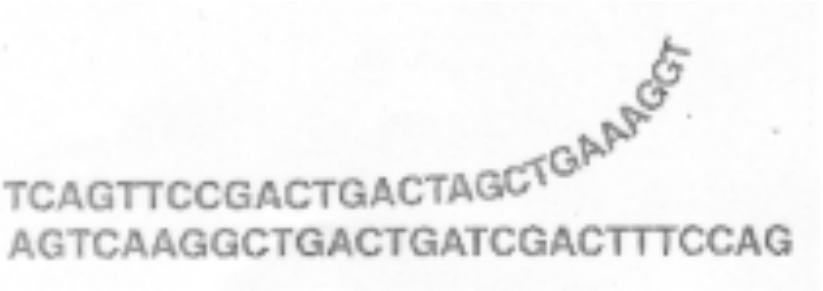


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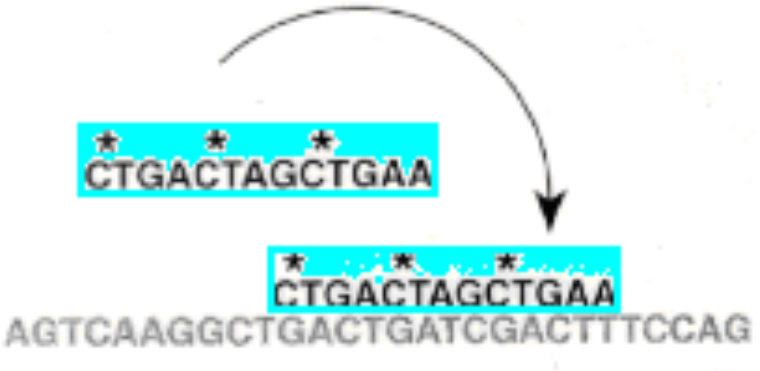
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Hybridization

- Denaturing is a process by which the hydrogen bonds of the original double-stranded DNA are broken, leaving a single strand of DNA whose bases are available for hydrogen bonding.



- Once the DNA has been denatured, a single-stranded radioactive probe [light blue] can be used to see if the denatured DNA contains a sequence similar to that on the probe.



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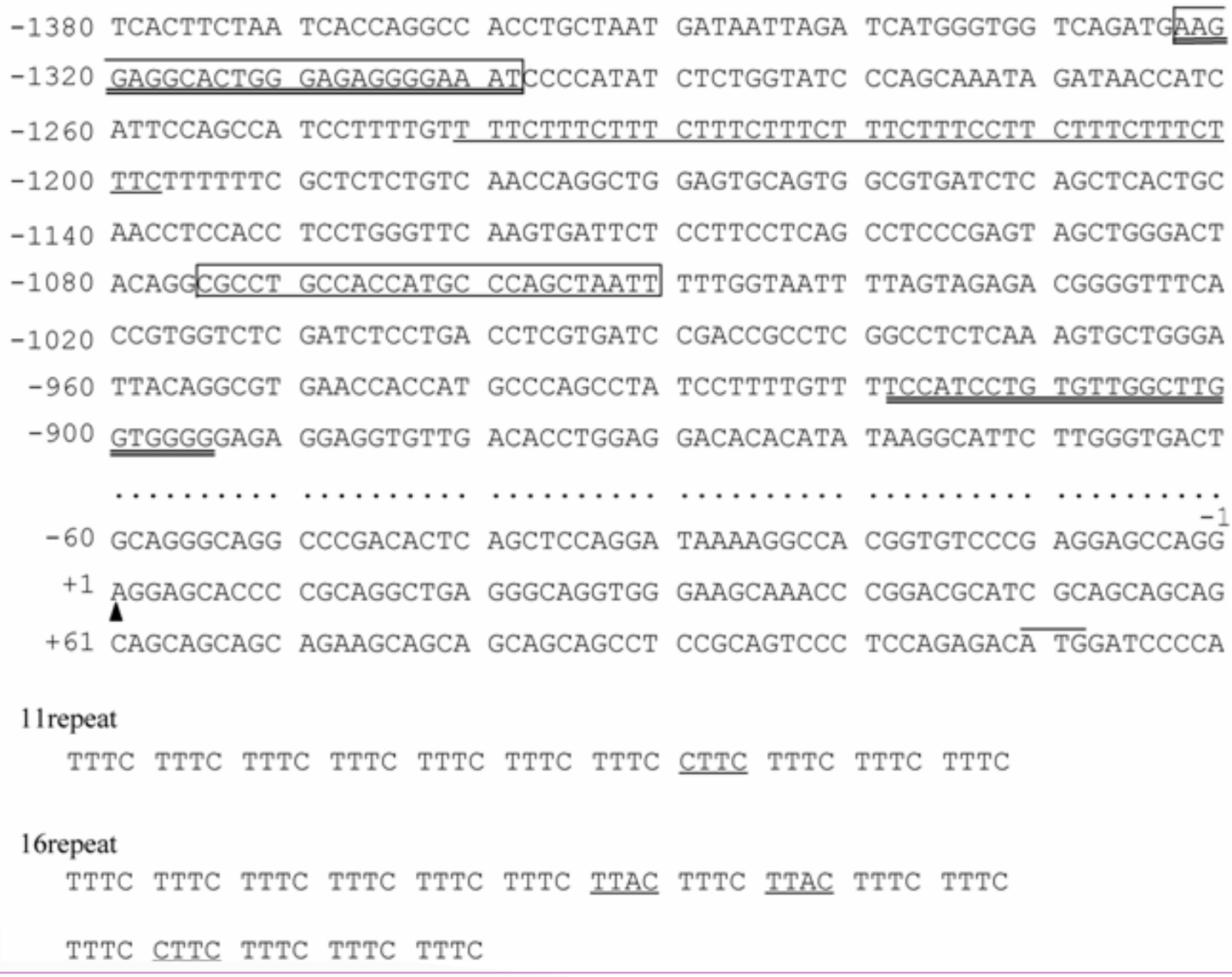
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VNTRs

- Every strand of DNA has pieces that contain genetic information which informs an organism's development **(exons)** and pieces that, apparently, supply no relevant genetic information at all **(introns)**.
- Although the introns may seem useless, it has been found that they contain repeated sequences of base pairs.
- These sequences, called Variable Number Tandem Repeats (VNTRs), can contain anywhere from twenty to one hundred base pairs.

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11repeat

TTTC TTTC TTTC TTTC TTTC TTTC TTTC CTTC TTTC TTTC TTTC

16repeat

TTTC TTTC TTTC TTTC TTTC TTTC TTAC TTTC TTAC TTTC TTTC

TTTC CTTC TTTC TTTC TTTC

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VNTRs

- **VNTRs are hereditary and hence unique**
- Hence an individual will have VNTRs inherited from the mother or father, or a combination of both, but never a VNTR either of the parents do not have.
- The more VNTR probes used to analyze a person's VNTR pattern, the more distinctive and individualized that pattern, or DNA fingerprint, will be.
- US CODIS - Combined DNA Information System – Uses 13 STR Loci

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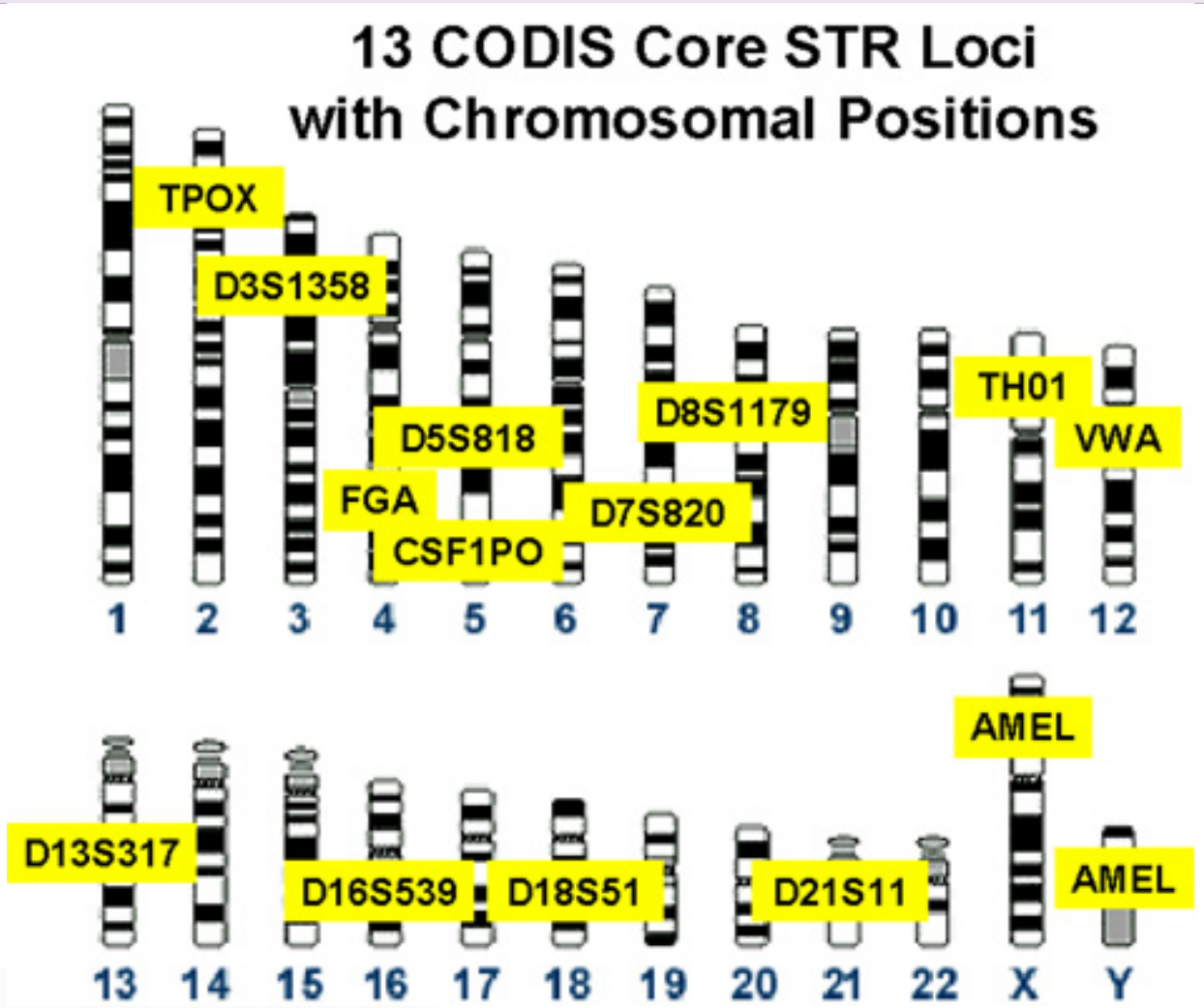
VNTRs

- Every human being has some VNTRs.
- To determine if a person has a particular VNTR, a Southern Blot is performed, and then the Southern Blot is probed, through a hybridization reaction, with a radioactive version of the VNTR in question.
- The pattern which results from this process is what is often referred to as a DNA fingerprint.

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CODIS



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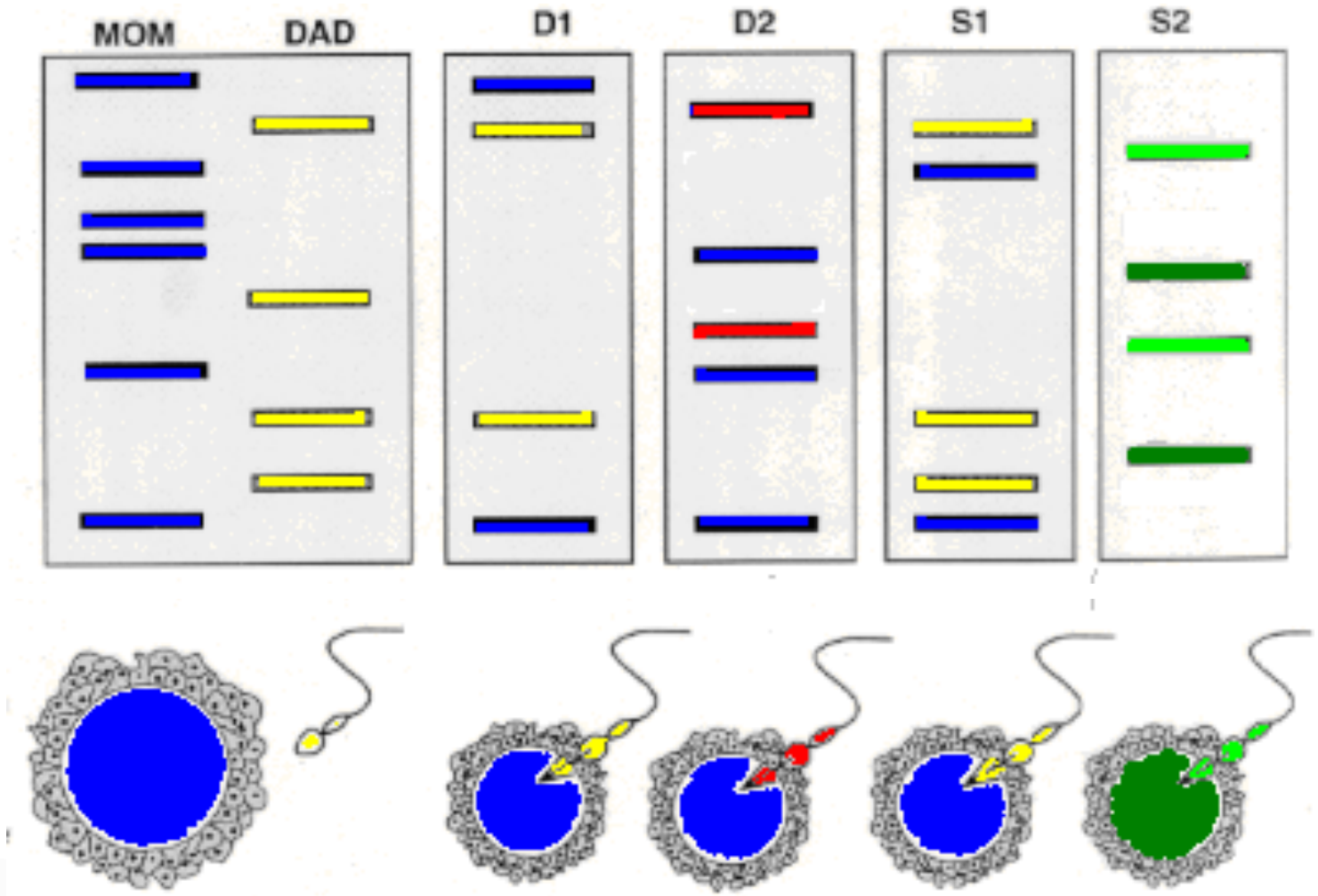
Practical Applications

- Paternity and Maternity
- Criminal Investigations and Forensics
- Personal Identification

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DNA Fingerprints



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